

Operations Research & Applications programming

Ch. 4

Duality and Post – Optimal Analysis

Introduction

- *In this Chapter*, From given a LPP (called the primal problem), we shall associate another LPP called the *Dual Problem* of the original (primal) problem.
- We shall see that the Optimal values of the primal and dual are the same provided both have *finite feasible solutions*. This topic is further used to develop another method of solving LPPs and it is also used in the sensitivity (or post-optimal) analysis.

Definition of the Dual Problem

- Given the primal problem (in standard form)

$$\text{Maximum } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

Subject to :

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

.

.

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$x_1, x_2, \dots, x_n \geq 0, b_1, b_2, \dots, b_m \geq 0$$

Definition of the Dual Problem

- The Dual problem is the LLP

$$\text{Minimum } w = b_1 y_1 + b_2 y_2 + \dots + b_m y_m$$

Subject to :

$$a_{11} y_1 + a_{21} y_2 + \dots + a_{m1} y_m \geq c_1$$

$$a_{12} y_1 + a_{22} y_2 + \dots + a_{m2} y_m \geq c_2$$

.

.

$$a_{1n} y_1 + a_{2n} y_2 + \dots + a_{mn} y_m \geq c_n$$

$y_1, y_2, \dots, y_n$ unrestricted in sign

Definition of the Dual Problem

- If the primal problem (in standard form) is

$$\text{Minimum } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

Subject to :

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

.

.

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$x_1, x_2, \dots, x_n \geq 0, b_1, b_2, \dots, b_m \geq 0$$

Definition of the Dual Problem

- Then the dual problem is the LLP

$$\text{Maximum } w = b_1y_1 + b_2y_2 + \dots + b_my_m$$

Subject to :

$$a_{11}y_1 + a_{21}y_2 + \dots + a_{m1}y_m \leq c_1$$

$$a_{12}y_1 + a_{22}y_2 + \dots + a_{m2}y_m \leq c_2$$

.

.

$$a_{1n}y_1 + a_{2n}y_2 + \dots + a_{mn}y_m \leq c_n$$

$$y_1, y_2, \dots, y_n \text{ unrestricted in sign}$$

Definition of the Dual Problem

□ When we constructing dual we note the following:

1. In the **dual**, there are as many (decision) **variables** as there are **constraints** in the **primal**.

We usually say y_i is the dual variable associated with the i^{th} constraint of the primal.

2. There are as many **constraints** in the **dual** as there are **variables** in the **primal**.

Definition of the Dual Problem

□ When we constructing dual we note the following:

3. If the primal is **maximum** then the dual is **minimum** and all constraints are $\geq$

If the primal is **minimum** then the dual is **maximum** and all constraints are $\leq$

4. In the **primal**, all variables are ≥ 0 while in the **dual** all the variables are **unrestricted in sign**.

Definition of the Dual Problem

□ When we constructing dual we note the following:

5. The objective function coefficients c_j of the primal are the RHS constants of the dual constraints.
6. The RHS constants b_i of the primal constraints are the objective function coefficients of the dual.
7. The coefficient matrix of the constraints of the dual is the transpose of the coefficient matrix of the constraints of the primal.

Problem 4(a) - Set 4.1A Page 155

❑ Write the dual of the primal LPP

$$\text{Maximize } z = -5x_1 + 2x_2$$

❑ Subject to the constraints :

$$-x_1 + x_2 \leq -2$$

$$2x_1 + 3x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

❑ Thus the primal in standard form is:

Problem 4(a) - Set 4.1A Page 155

$$\text{Maximize } z = -5x_1 + 2x_2 + 0x_3 + 0x_4$$

□ Subject to the constraint :

$$x_1 - x_2 - x_3 = 2$$

$$2x_1 + 3x_2 + x_4 = 5$$

$$x_1, x_2, x_3, x_4 \geq 0$$

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□ Hence the dual is:

$$\text{Minimize } w = 2y_1 + 5y_2$$

Subject to the constraint

$$\begin{array}{rcl} y_1 + 2y_2 & \geq & -5 \\ -y_1 + 3y_2 & \geq & 2 \\ -y_1 & \geq & 0 \\ y_2 & \geq & 0 \end{array} \left. \vphantom{\begin{array}{rcl} y_1 + 2y_2 & \geq & -5 \\ -y_1 + 3y_2 & \geq & 2 \\ -y_1 & \geq & 0 \\ y_2 & \geq & 0 \end{array}} \right\} \Rightarrow y_1 \leq 0, y_2 \geq 0$$

y_1, y_2 unrestricted in sign

Problem 4(b) - Set 4.1A Page 155

❑ Write the dual of the primal LPP

$$\text{Minimize } z = 6x_1 + 3x_2$$

❑ Subject to the constraints :

$$6x_1 - 3x_2 + x_3 \geq 2$$

$$3x_1 + 4x_2 + x_3 \geq 5$$

$$x_1, x_2, x_3 \geq 0$$

❑ Thus the primal in standard form is:

Problem 4(b) - Set 4.1A Page 155

$$\text{Minimize } z = 6x_1 + 3x_2 + 0x_3 + 0x_4 + 0x_5$$

□ Subject to the constraint :

$$6x_1 - 3x_2 + x_3 - x_4 = 2$$

$$3x_1 + 4x_2 + x_3 - x_5 = 5$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

Problem 4(b) - Set 4.1A Page 155

□ Hence the dual is: *Maximize* $w = 2y_1 + 5y_2$

Subject to the constraint

$$6y_1 + 3y_2 \leq 6$$

$$-3y_1 + 4y_2 \leq 3$$

$$y_1 + y_2 \leq 0$$

$$\left. \begin{array}{rcl} -y_1 & \leq & 0 \\ -y_2 & \leq & 0 \end{array} \right\} \Rightarrow y_1 \geq 0, y_2 \geq 0$$

y_1, y_2 unrestricted in sign

Problem 4(c) - Set 4.1A Page 155

□ Write the dual of the primal LPP

$$\text{Maximize } z = x_1 + x_2$$

□ Subject to the constraints :

$$2x_1 + x_2 = 5$$

$$3x_1 - x_2 = 6$$

$$x_1, x_2 \text{ unrestricted}$$

□ Thus the primal in standard form is:

Problem 4(a) - Set 4.1A Page 155

$$\text{Maximize } z = x_1^+ - x_1^- + x_2^+ - x_2^-$$

Subject to the constraint :

$$2x_1^+ - 2x_1^- + x_2^+ - x_2^- = 5$$

$$3x_1^+ - 3x_1^- - x_2^+ + x_2^- = 6$$

$$x_1^+, x_1^-, x_2^+, x_2^- \geq 0$$

Problem 4(a) - Set 4.1A Page 155

□ Hence the dual is:

$$\text{Minimize } w = 5y_1 + 6y_2$$

Subject to the constraint

$$\left. \begin{array}{l} 2y_1 + 3y_2 \geq 1 \\ -2y_1 - 3y_2 \geq -1 \end{array} \right\} \Rightarrow 2y_1 + 3y_2 = 1$$

$$\left. \begin{array}{l} y_1 - y_2 \geq 1 \\ -y_1 + y_2 \geq -1 \end{array} \right\} \Rightarrow y_1 - y_2 = 1$$

y_1, y_2 unrestricted in sign

Summary of the Rules for Constructing the Dual

- The general conclusion from the preceding examples is that the variables and constraints in the primal and dual problems are defined by the rules in **Table 4.2** :

Maximization problem		Minimization problem
Constraints		Variables
$\geq$	$\leq$	≤ 0
$\leq$	$\geq$	≥ 0
$=$	$=$	Unrestricted
Variables		Constraints
≥ 0	$\leq$	$\geq$
≤ 0	$\geq$	$\leq$
Unrestricted	$=$	$=$